University of Florida, Department of Mathematics

Algebra PhD Exam, August 2025

Answer seven problems. (If you turn in more, the first seven will be graded.) Within reason, you may quote theorems as long as you state them clearly.

Note. Below ring means associative ring with identity, and module means unital module unless otherwise specified.

1. Let K be a field and let $f(x) \in K[x]$ be a non-zero polynomial.

(a) Define what is a splitting field for $f(x)$ over K.

(b) If F_1 and F_2 are splitting fields for $f(x)$ over K prove, from your definition, that F_1 and F_2 are isomorphic fields.

2. Suppose that p is an odd prime and the field K is the extension of $\mathbb{Q}$ generated by a primitive p-th root of 1 in $\mathbb{C}$.

(a) Show that K contains a unique quadratic extension F of $\mathbb{Q}$.

(b) What is $\mathrm{Gal}(K/F)$? Justify your answer in detail.

3. Let A and B be finite abelian groups and suppose that $|A|$ and $|B|$ are relatively prime. Let $T = A \otimes_{\mathbb{Z}} B$. Show that $|T| = 1$.

4. Let $\mathcal{C}$ be a concrete category.

(a) Define what is meant by a free object in $\mathcal{C}$.

(b) Let $\mathcal{G}$ be the category of all finite groups. Prove that the free objects in $\mathcal{G}$ are exactly the trivial groups.

5. Let R be a ring with identity. Prove that the following two conditions are equivalent for a left R-module P. (You may not assume any properties of projective modules).

1. Every short exact sequence of R-modules

$$0 \to A \to B \to P \to 0$$

splits;

2. There is a free R-module F and an R-module K such that $F \simeq K \oplus P$.

6. Let K be a field and $K(x)$ the field of rational functions over K. Let $\frac{f}{g} \in K(x)$, with $\frac{f}{g} \notin K$, where f and g are relatively prime elements of $K[x]$.

(a) Show that $\frac{f}{g}$ is transcendental over K.

(b) Prove that $K(x)$ is a finite extension of $K\left(\frac{f}{g}\right)$.

7. State and prove the Lying-Over Theorem on prime ideals and ring extensions.

8. Prove that an invertible ideal in an integral domain that is a local ring is principal.

9. Show that for any extension $A \subseteq B$ of commutative rings with 1, any A-module M and any B-module N, there is an isomorphism $\mathrm{Hom}_B(M \otimes_A B, N) \simeq \mathrm{Hom}_A(M, N_A)$ (here N_A denotes N considered as an A-module by means of the ring inclusion homomorphism $A \to B$).

10. Let R be a commutative ring with 1 with $1 \neq 0$. Prove that the set consisting of 0 and all zero divisors in R contains at least one prime ideal of R.

11. Classify (up to ring isomorphism) all (not necessarily commutative) semisimple rings of order 240.